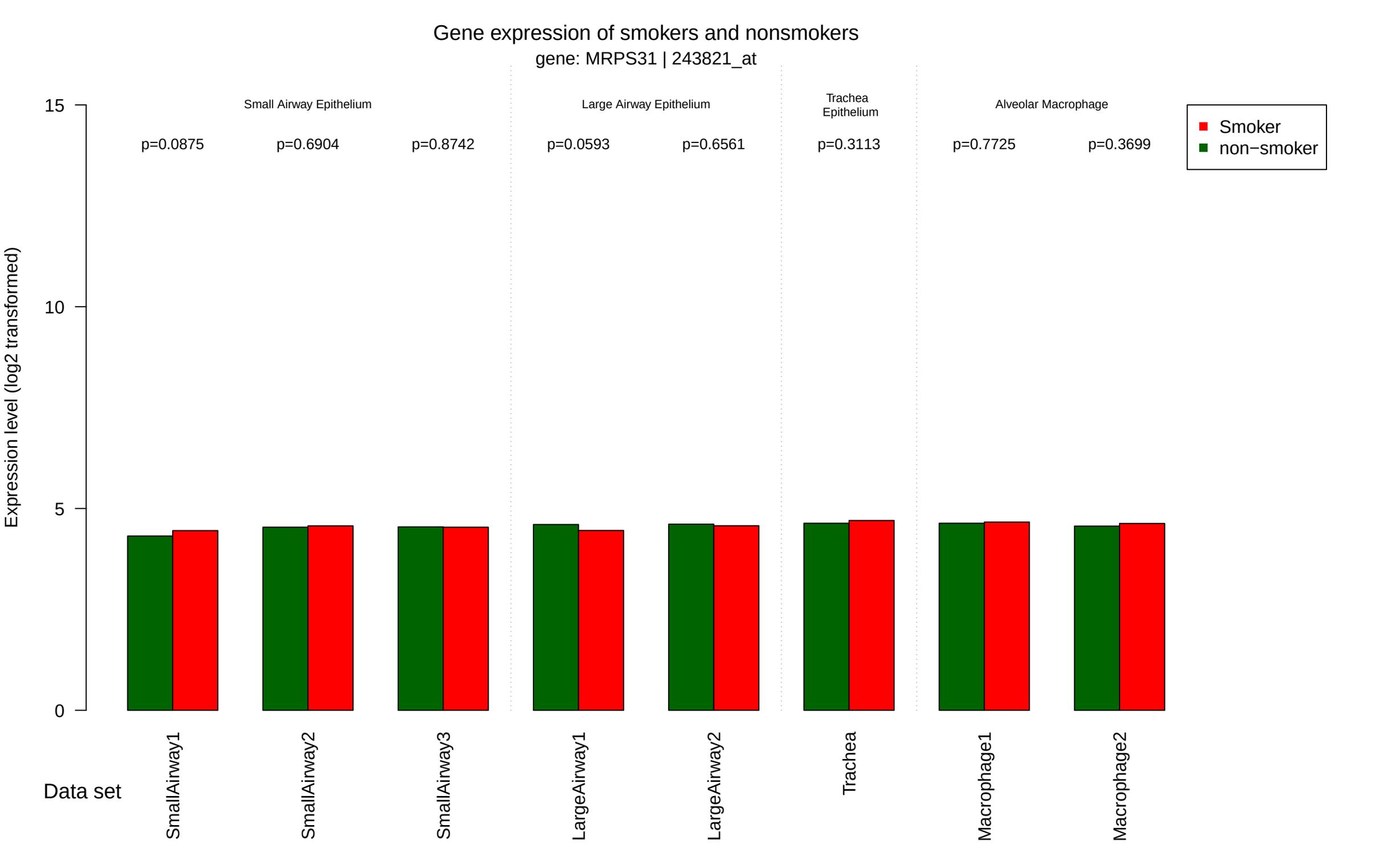
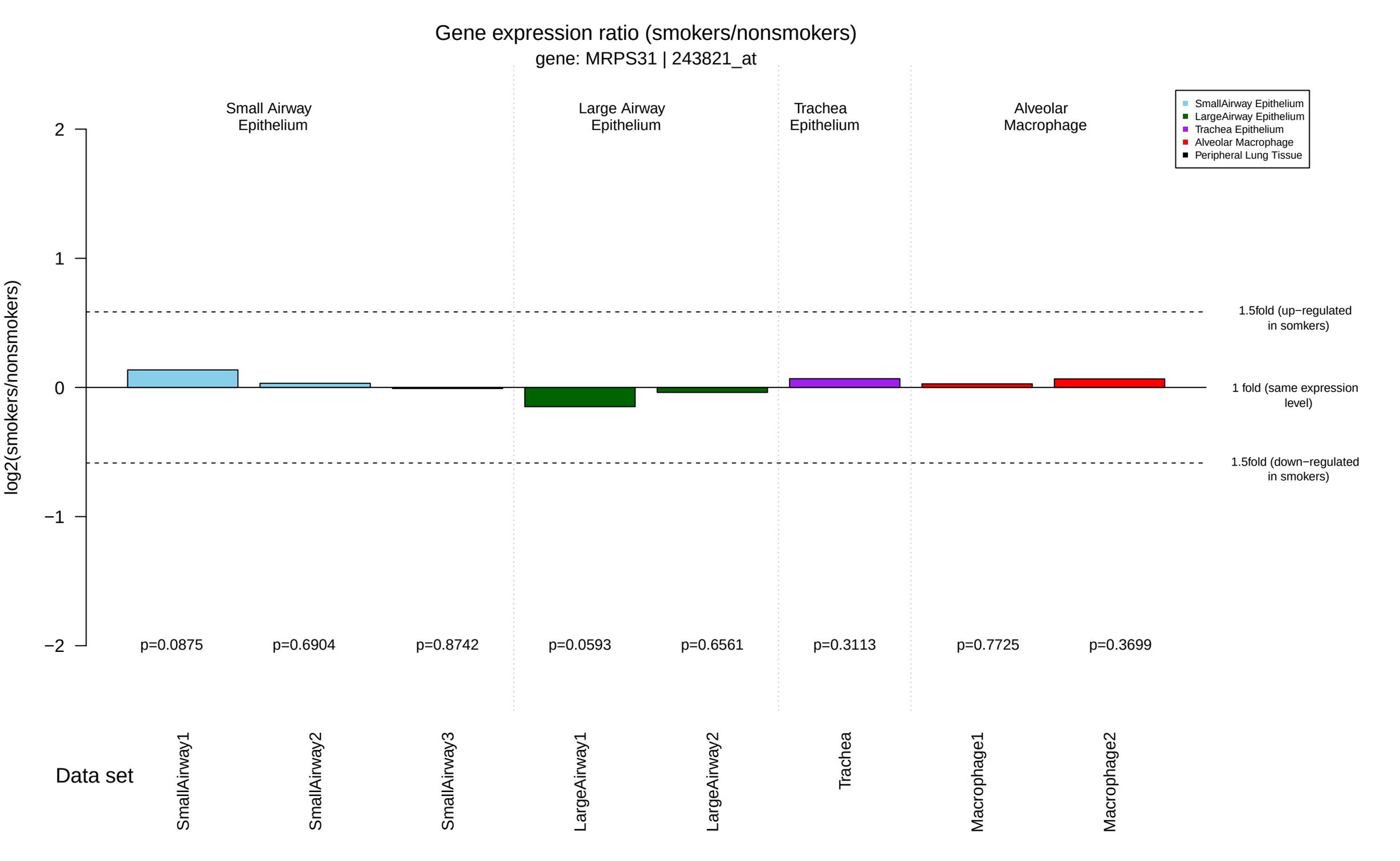
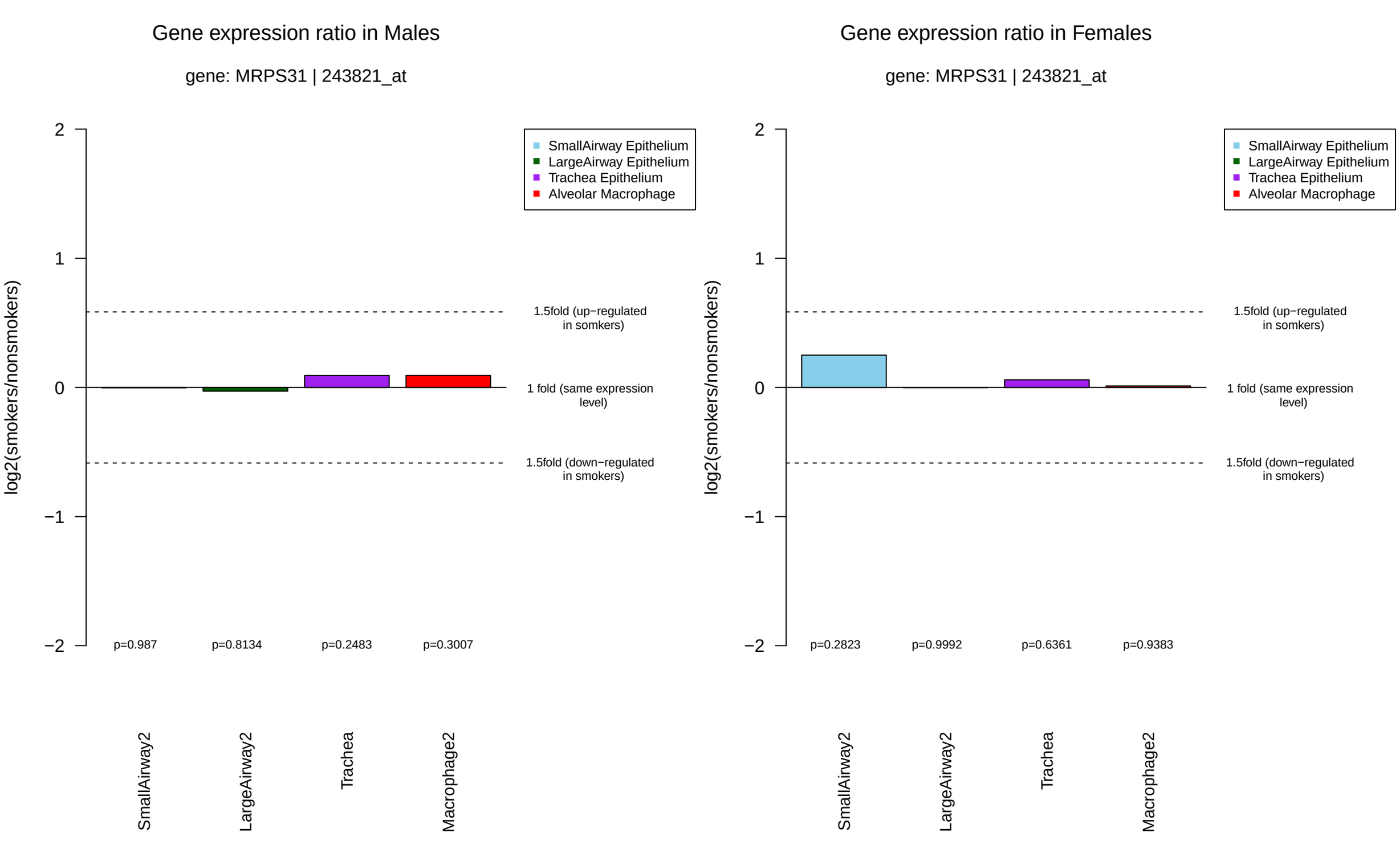
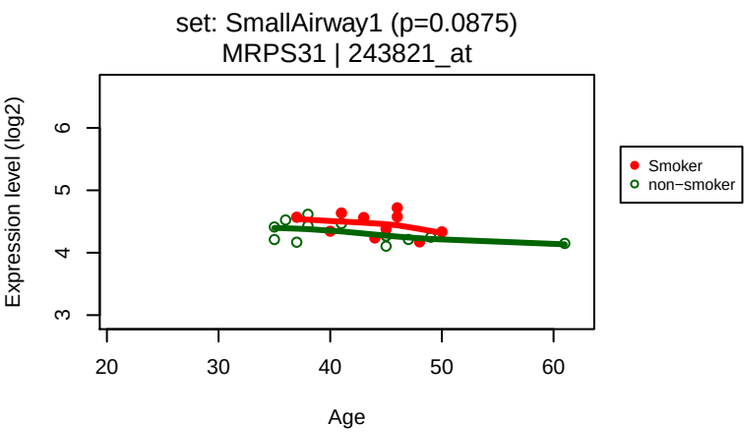
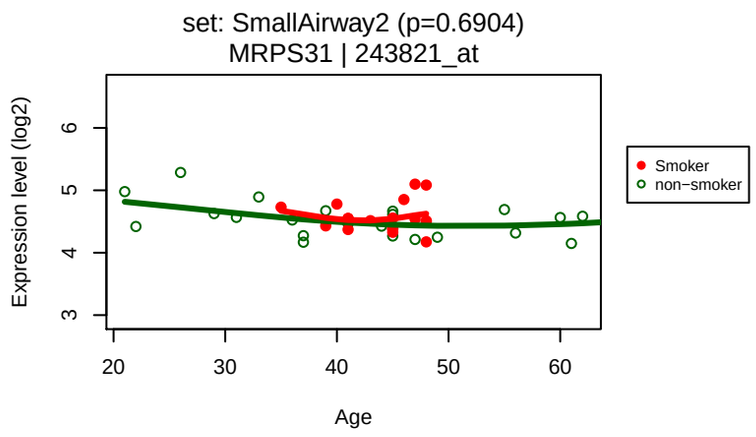
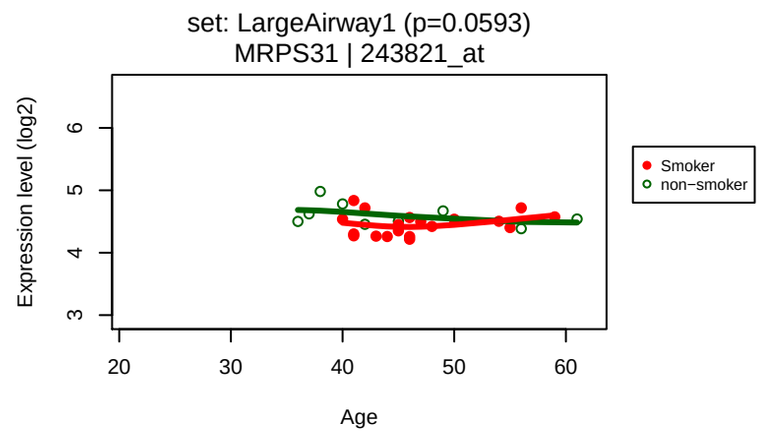

Gene expression of smokers and nonsmokers
gene: MRPS31 | 243821_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0875
p=0.6904
p=0.8742
p=0.0593
p=0.6561
p=0.3113
p=0.7725
p=0.3699
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

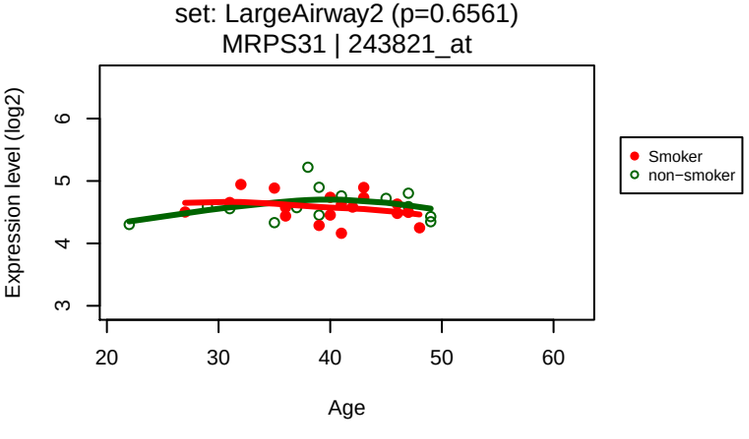

Gene expression ratio (smokers/nonsmokers)
gene: MRPS31 | 243821_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0875
p=0.6904
p=0.8742
p=0.0593
p=0.6561
p=0.3113
p=0.7725
p=0.3699
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

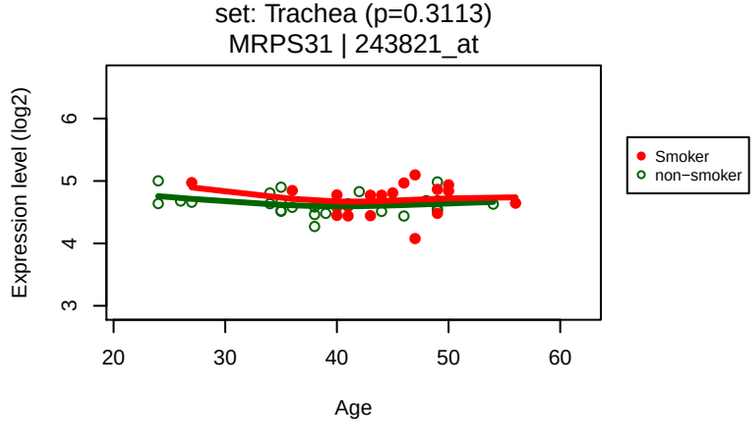

Gene expression ratio in Males
gene: MRPS31 | 243821_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.987
p=0.8134
p=0.2483
p=0.3007
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRPS31 | 243821_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2823
p=0.9992
p=0.6361
p=0.9383
SmallAirway2
LargeAirway2
Trachea
Macrophage2

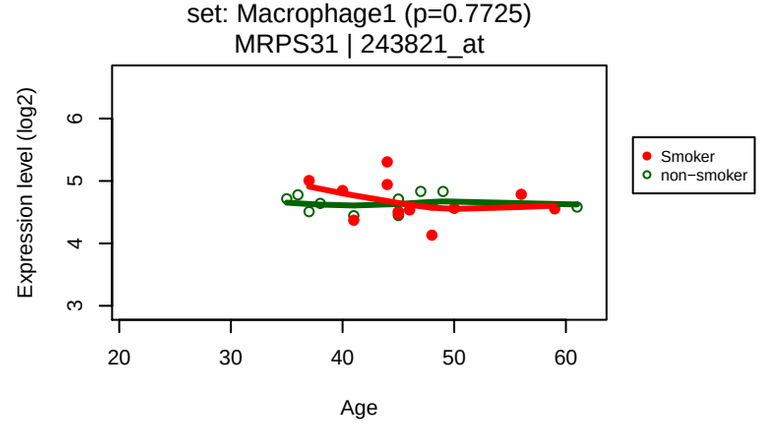

set: SmallAirway1 (p=0.0875)
MRPS31 | 243821_at
Expression level (log2)
Age
Smoker
non−smoker

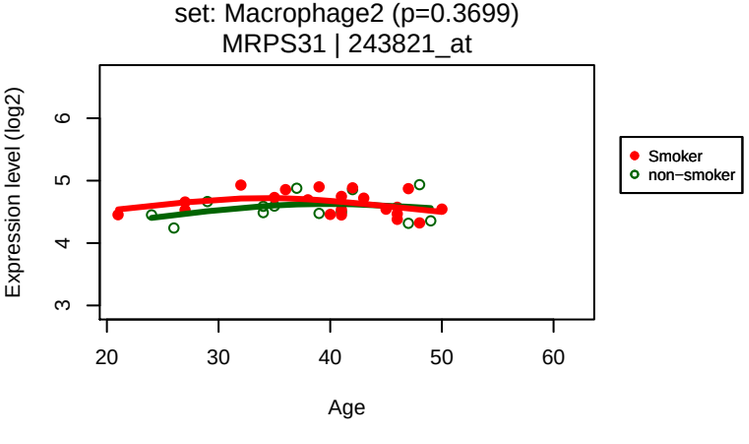

set: SmallAirway2 (p=0.6904)
MRPS31 | 243821_at
Expression level (log2)
Age
Smoker
non−smoker

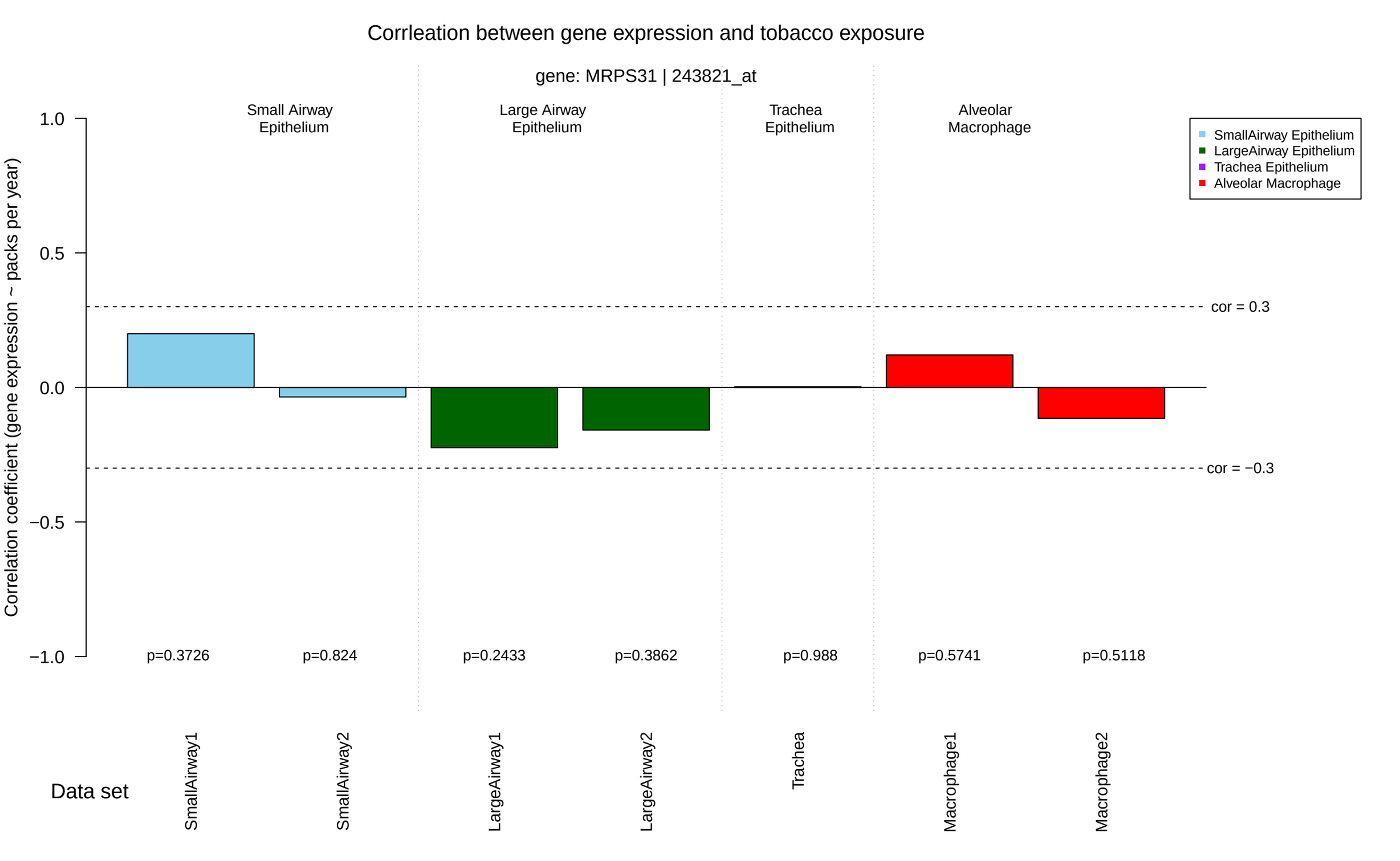

set: LargeAirway1 (p=0.0593)
MRPS31 | 243821_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6561)
MRPS31 | 243821_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3113)
MRPS31 | 243821_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7725)
MRPS31 | 243821_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3699)
MRPS31 | 243821_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRPS31 | 243821_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3726
p=0.824
p=0.2433
p=0.3862
p=0.988
p=0.5741
p=0.5118
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2